

AudioEval: Music Score-based Instrument Performance Assessment System

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Abstract

Useful practice feedback identifies an error, locates it in the score, and lets the student replay the relevant passage. We present AudioEval, a layered approach combining transcription, score alignment, and error labelling. Transcription estimates the notes in a recording; score alignment associates them with the intended written notes. The current AudioEval uses a dual-resolution CTC transcriber, retrieves the practised passage, and aligns candidate notes using bounded repetition and ornament templates. Acoustic verification supports note-error decisions, and uncertain alignments are explicitly withheld. Synthetic performances supply training examples with known relationships between played and written notes. AudioEval, using the dual-resolution CTC transcriber and passage aligner, achieved a score-based weighted micro-F1 of 61.04% on 358 synthetic validation recordings, compared with 41.39% for Polytune and 34.71% for LadderSym. This measure assesses both score correspondence and event type across correct, repeated, and erroneous notes. On 11 manually labelled clarinet recordings, AudioEval achieved error-event micro-F1 of 25.49%. Two case studies illustrate the current pipeline’s alignment and feedback. We also present MusicPal, an implemented desktop practice companion that delivers score-based audiovisual feedback and retains practice sessions. Making it available to more players offers a route to expand the limited collection of real clarinet practice data through human review and subsequent model improvement.

1 Introduction

A student practising a musical instrument alone needs to know which notes require attention and where to find them in the score. A useful feedback system should identify the type of error, highlight its location, and let the student replay the corresponding passage.

Recent error-detection methods learn to compare a performance with its intended reference. Polytune and LadderSym use synthetic musical audio and scores to train this comparison at a scale that would be difficult to obtain through manual annotation [4, 5]. LadderSym also reports evaluation on real beginner piano performances.

Their output categories are correct, missed, and extra notes [4, 5]. A return to an earlier passage can therefore appear as a sequence of extra notes without identifying which passage was repeated. When a student returns to an earlier passage, the same written notes occur again at different times in the recording. Interpreting such performances requires acoustic evidence to be considered in its musical context: identifying what was played is closely connected to determining where it belongs in the score.

AudioEval combines transcription, score alignment, and error labelling to make this connection. Its current acoustic network learns clarinet note sequences from dual-resolution spectrograms while retaining uncertain candidates. A passage locator connects partial recordings to a full score, and a bounded dynamic program compares repetition- and ornament-aware templates using pitch, confidence, and onset timing. Separate acoustic checks verify possible extra and

missed notes. The resulting feedback identifies affected score notes and replay intervals, with an explicit uncertain state when alignment is unreliable.

AudioEval achieved a score-based weighted micro-F1 of 61.04% on 358 synthetic validation recordings, compared with 41.39% for Polytune and 34.71% for LadderSym. The comparison uses common recordings, reference annotations, and a scoring rule that rewards both correct location and correct event type. Section 5 describes the measure and the training conditions of the compared systems. The AudioEval configuration combines the CTC transcriber and passage aligner described in Section 4. These are end-to-end feedback results, rather than measurements of the transcriber alone.

We also built MusicPal, a working desktop practice companion for recording performances and reviewing AudioEval’s feedback through score animation and audio playback. Its saved-session workflow offers a route to collect real clarinet practice data, which remain limited in our current collection. With user permission and human review, these recordings can support a data flywheel in which continued use supplies training examples and improved models return better feedback to players.

The contributions are:

1. AudioEval, a layered method combining clarinet transcription, bounded passage alignment, explicit repetition tracking, and verified score-linked feedback, together with a comparison against two existing detectors.
2. An implemented MusicPal practice companion that presents score-based audiovisual feedback and retains sessions, together with a plan for using reviewed practice data to improve subsequent models.

2 Related Work

2.1 From Transcription to Practice Feedback

Automatic music transcription estimates which notes are present in a recording and when they occur. Models such as Onsets and Frames, MT3, and Basic Pitch provide different approaches to this task [2, 7, 9]. Practice feedback requires an additional reference: the score the player intended to perform. The same detected pitch can be correct in one passage and a wrong note in another. Locating the sound in the score is therefore part of interpreting the performance.

Score-informed detectors combine acoustic evidence, alignment, and note comparison [1, 15]. Dynamic time warping aligns two sequences while accommodating changes in speed [6]. More specialised approaches address local performance errors, repeats, and skips [12, 13]. RUMAA brings transcription, score alignment, and mistake detection into a unified framework with repeat handling [3]. The current AudioEval combines a clarinet-specific CTC transcriber with passage retrieval and bounded repeat templates, retaining alternative note candidates during alignment.

2.2 Learned Error Detection

Polytune learns to compare practice audio with audio rendered from the reference score and predicts correct, missed, and extra notes [4]. LadderSym extends this approach with repeated interaction between the two input streams and a symbolic score prompt [5]. Both systems learn to predict error labels from the practice and reference inputs.

AudioEval represents the correspondence between played and written notes explicitly. Repeat templates preserve canonical score identity and performed occurrence; the current feedback exporter reports passage repetition separately from first-pass content errors. The comparison with Polytune and LadderSym evaluates AudioEval as a complete system, including its CTC transcriber, passage aligner, and error verification. Because their original outputs describe

timestamped notes, a common conversion assigns their predictions to score locations for the evaluation used here.

2.3 Synthetic and Real Performance Data

Large transcription datasets provide detailed note annotations: MAESTRO pairs piano audio with MIDI, while MusicNet and URMP cover additional instrumental settings [10, 11, 14]. Such annotations describe the notes that were played. Error detection also needs the intended score and an account of deviations from it.

Synthetic generation supplies these relationships by modifying a score before rendering its performance as audio. Polytune and LadderSym use this approach for error supervision, including instruments represented in CocoChorales [4, 5, 16]. Their synthetic datasets support controlled comparisons with known errors. LadderSym additionally evaluates on a collection of 20 real beginner piano recordings [5]. Our synthetic generation preserves the score location of each altered and repeated note to supervise score-linked clarinet practice feedback. MusicPal now provides a working interface for practice feedback and session retention; extending its use to more players offers a route to collect real clarinet recordings for manual annotation.

3 Synthetic Training Data and Design Rationale

3.1 Synthetic Training Performances

We generate labelled training performances by introducing wrong notes, missed notes, extra notes, rhythm changes, and unintended repetitions into a reference score. The modified performance is then rendered with a clarinet SoundFont, a sampled-instrument sound library. During generation, we record how each performed event relates to the original score.

For example, a wrong note changes the played pitch while preserving the intended written location. A missed note remains a target in the score although it produces no sound. An extra note is recorded at its position in the generated performance but has no written counterpart. When a passage is repeated, each performed occurrence retains its link to the original written notes. These relationships provide training targets for both score alignment and error labelling.

The selected acoustic checkpoint is trained using realistic, fast-note, and practice-like synthetic families, with the final fine-tuning configuration recorded in Appendix A.1. Pitch conventions, note timing, and correspondences between generated notes and the reference score are checked before use. During inference, the detector receives only the recording and reference score; generation records supply training targets and evaluation annotations. Section 5.1 reports the evaluation cohort and the separate baseline training conditions.

3.2 Design Rationale

An error label is useful during practice when the student can locate the note in the score and replay the corresponding sound. AudioEval therefore keeps track of score locations and recording intervals as it identifies note errors and unintended repetitions.

We train a clarinet-specific transcriber with complementary time resolutions to represent both pitch structure and short attacks. AudioEval retains uncertain note candidates for score-informed selection, retrieves the attempted passage, and compares bounded restart hypotheses. Separate error-verification models help distinguish a performance mistake from a transcription failure. Table 1 summarises these choices and the alternatives considered.

Table 1: Design choices in AudioEval and MusicPal, their alternatives, and their intended purposes.

Design aspect	Alternatives	Selected approach and purpose
Acoustic recognition	Train a task-specific transcriber; reuse a pretrained transcriber	Train a dual-resolution CTC network with auxiliary pitch and boundary supervision.
Note selection	Commit to one transcription; consider alternatives during alignment	Retain primary, optional, and alternative-pitch candidates for structured score alignment.
Score alignment	Require forward progress; allow returns to earlier passages	Retrieve a bounded passage and compare repeat and ornament templates with timing refinement.
Error decisions	Predict all labels directly; interpret a shared correspondence	Verify extra and missed proposals acoustically; report canonical note identities and uncertainty.
Device processing	Run analysis on the device; use a remote server	Use the device for recording and presentation and the server for AudioEval.

4 AudioEval

AudioEval takes a monophonic clarinet recording and a MusicXML score and returns score-linked wrong-note, missed-note, extra-note, and unintended-repetition labels. The current implementation combines a trainable acoustic transcriber with a structured score aligner: acoustic evidence proposes what was played, and the aligner determines which score events explain it while allowing genuine departures from the notation. Figure 1 summarises the inference path.

This section describes the dual-resolution CTC transcriber and bounded passage aligner used by AudioEval. The quantitative configuration is the passage-enabled v9 pipeline documented in Appendix A.1; the v10 boundary refinement described below is a separate experimental extension.

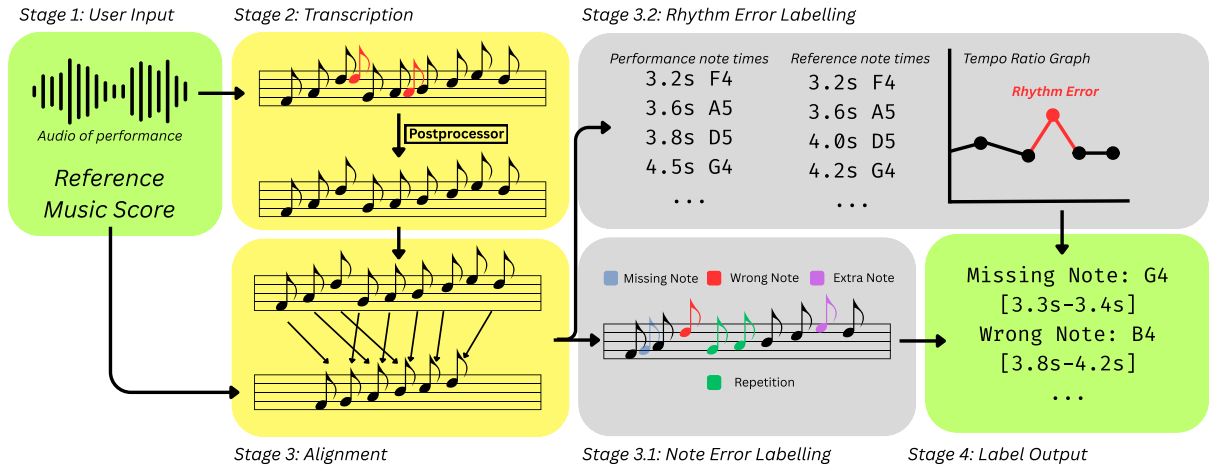


Figure 1: AudioEval methodology overview: performance audio is transcribed and postprocessed, aligned to the reference score, and converted into score-linked note-error and repetition feedback with recording intervals. In the current implementation, transcription uses the dual-resolution CTC model and alignment uses the bounded passage pipeline. The rhythm-error branch shown in the schematic is not enabled in the current v9 configuration.

4.1 Score Representation and Acoustic Input

We parse the intended part into a canonical sequence $S = (s_1, \dots, s_M)$, where each event stores its written MIDI pitch, onset and duration in quarter-note units, measure, part, and unique score index. Tied notes form one sustained event; separately articulated notes retain separate identities even when their pitches agree. These identities distinguish two occurrences of the same melody in different score locations. Ornament annotations are retained for the aligner’s local performance templates.

Audio is converted to mono at 22,050 Hz. The current acoustic model predicts written pitches for B-flat clarinet: sounding pitch is two semitones below written pitch. Training loaders use the recorded pitch-space metadata; the deployed configuration assumes a written B-flat-clarinet score. Concert-pitch inputs therefore require an explicit conversion. Inference uses the reference MusicXML and recording, without access to the modified synthetic score or the generation-time note correspondence.

4.2 Dual-Resolution Acoustic Transcription

The transcriber is a clarinet-specific neural network trained on synthetic performances. A long analysis window preserves pitch structure, while a short window retains attacks and brief notes. Both log-mel spectrograms use a 256-sample hop, approximately 11.61 ms. The long branch has a 2,048-sample window and 128 mel bands; the short branch has a 512-sample window, a 1,024-point Fourier transform, and 64 mel bands. Each branch is normalised using recording-level quantiles. Their frames share the same time origin.

Separate convolutional encoders compress the frequency axis while preserving time resolution. Their outputs are concatenated, projected to 256 channels, and added to a projection of the original 192 mel bands. Twelve residual temporal convolution blocks use kernel width five and dilations 1, 2, 4, 8, 16, 32, repeated twice. This supplies approximately 2.9 seconds of convolutional context on each side of a frame. The network outputs a connectionist temporal classification (CTC) distribution over written MIDI pitches 52–100 and a blank symbol, together with auxiliary estimates of pitch, voicing, onset, boundary, rearticulation, and confidence. Long recordings are processed in overlapping windows whose probabilities are averaged with tapered weights.

CTC [8] represents an ordered note sequence without requiring each output note to occupy its full acoustic duration. For a target pitch sequence Y and acoustic features X , the sequence objective is

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi: \mathcal{B}(\pi)=Y} \prod_{t=1}^T P_{\theta}(\pi_t | X), \quad (1)$$

where $\mathcal{B}$ collapses consecutive identical tokens and removes blanks. Distinct successive notes of the same pitch require a separating blank in the CTC path. The auxiliary boundary targets provide additional supervision for these rearticulations.

Decoding scales the blank probability before taking framewise maxima and collapsing runs. Each primary candidate retains its pitch, onset, peak CTC support, and an alternative pitch with its support. Additional non-blank probability peaks in otherwise uncovered regions form optional candidates. Brief intermediate pitches with weak onset and rearticulation evidence can also be marked optional. The aligner may retain an optional candidate if the score supports it, or discard it; short duration alone does not remove a note. Candidate end times are initially estimated from the next onset, so they are playback approximations rather than independently measured offsets.

4.3 Adjacent Same-Pitch Boundaries

A sustained note can produce two CTC emissions, while a genuine repeated note can have little silence between attacks. The v9 postprocessor examines adjacent equal-pitch candidates using local energy and spectral continuity. A separate waveform envelope uses centred 4 ms RMS windows at 1 ms intervals, allowing the gap check to inspect interruptions shorter than the mel hop. A detected low-energy core with stronger flanks prevents merging; missing or insufficient evidence also retains the boundary. The permissive v9 continuity settings can nevertheless merge real rearticulations. Absence of a detected gap is not evidence that the performer played only one note.

The experimental v10 extension addresses a restricted subset of these cases. After an initial alignment, it considers a merged group only when its original, confident CTC candidates correspond to an equal number of consecutive, distinct, same-pitch score events. A small random forest combines gap, energy, spectral, attack, and voicing features to vote on whether to restore an original boundary. It does not invent an onset. The proposed transcription is realigned and accepted only if the restored candidates map to separate consecutive score events and mappings, deletions, and error identities outside the affected groups remain unchanged. Otherwise the original result is returned. This check limits changes outside the proposed repair; it does not prove that a restored boundary is acoustically correct.

4.4 Locating a Practised Passage

A recording may cover only part of a long score. We first retrieve plausible passages using confidence-weighted subsequence edit distance between reliable transcribed pitches and the canonical score. Unmatched score prefixes and suffixes have no cost, so notes outside the attempted passage are not automatically treated as omissions. Two rolling dynamic-programming rows keep retrieval memory linear in score length.

The locator retains up to four distinct passages and adds eight context notes at each end. Each window is then evaluated by the detailed aligner below. Its ranking combines alignment cost per transcribed candidate with retrieval dissimilarity. When several locations remain close to the best candidate and pass the alignment-quality threshold, the current runtime selects the earliest plausible location and records the ambiguity. This is a deterministic convention for equivalent passages, not proof of the performer’s intended location. Short supplied excerpts bypass retrieval. Insufficient evidence or a search exceeding the configured limits produces an uncertain result.

4.5 Repetition- and Ornament-Aware Alignment

For each retrieved passage, the aligner constructs a bounded set of performance templates. Linked units retain their original score index and an occurrence index; additional ornament units carry their expected pitch and occurrence without inventing a canonical score note. The no-restart template competes with templates containing one or two additional copies of a passage. Local proposals come from consecutive repeated phrases in the transcription and can represent two non-overlapping restart regions. Measure-aligned repeat proposals fill the remaining search budget, prioritised by agreement with the observed sequence length. Written ornament patterns allow supported embellishment notes and linked score units to be considered together. The search does not cover arbitrary nested repeats or unrestricted jumps.

For example, the written sequence 1, 2, 3, 4 and performance 1, 2, 3, 2, 3, 4 can be explained by a template that copies events 2–3 and then resumes at event 4. Both visits preserve the same canonical score identities, while occurrence indices distinguish their recording intervals. A repeated motif already present in the written sequence can instead be explained by the no-restart template.

For a template U_h , dynamic programming minimises edit costs over acoustic candidates C . Permitted operations include pitch match or substitution, insertion of an extra candidate, deletion of a score unit, rejection of an unreliable candidate, selection of a supported alternative pitch, and permitted ornament or same-pitch span transitions. A schematic recurrence is

$$D_h(i, j, q) = \min_{o \in \mathcal{O}(i, j, q)} [D_h(i - \Delta_i(o), j - \Delta_j(o), q_o) + c_o(C, U_h)], \quad (2)$$

where q is the operation state and c_o is the configured cost. The start state has cost $D_h(0, 0, 0) = 0$; unreachable states have infinite cost. The state preserves constraints on linked units and merged spans. Candidate confidence affects insertion and rejection costs, and an additional penalty discourages unnecessary repeat copies. These are explicit edit costs, not a learned neural path score.

The first pass selects the lowest-cost repeat template. Confident pitch matches then anchor a map from the template’s quarter-note positions to recording time. If at least four anchors are available, a second dynamic-programming pass on the selected template adds a capped onset-deviation cost,

$$c_{\text{time}}(i, j) = \lambda_t \min \left\{ \left(\frac{t_i - \hat{t}_j}{\tau_j} \right)^2, \kappa \right\}, \quad (3)$$

where $\hat{t}_j$ is the predicted onset and τ_j depends on expected note duration with a fixed lower bound. This pass helps resolve repeated-pitch ambiguities while allowing expressive timing. It refines the selected template rather than optimising timing jointly over every repeat hypothesis. Window-local indices are finally mapped back to the original full-score indices.

4.6 Error Verification and Score-Linked Feedback

The alignment proposes substitutions, insertions, deletions, and repeated passages. A pitch substitution on the first pass gives a wrong-note label. An unexplained retained candidate gives a possible extra note, and an unfilled first-pass score unit gives a possible missed note. Repeat-copy indices identify the repeated passage and its additional occurrences; the current feedback exporter does not issue separate content-error labels within replayed copies.

Extra and missed proposals pass through separate logistic classifiers using acoustic support, duration, neighbouring alignment, local timing, and structural context. Missed-note verification also queries a small convolutional network on the dual-resolution spectrogram around the expected onset, conditioned on harmonic templates for the expected pitch and its neighbours. It tests whether a note may have been played but omitted by the transcriber. Low-support error calls are withheld. Gate scores are operating-point scores and are not assumed to be calibrated error probabilities.

A clip-level check withholds all error labels when the fraction of alignment events marked as matches or repeat copies is too low. This heuristic includes replay events and is not a calibrated accuracy estimate. Retrieval failure and resource-limit failures also return an explicit uncertain status. Unplayed score regions and withheld decisions remain unassessed; an empty error list must not be interpreted as a verified clean performance.

Each emitted label carries canonical affected-note identities, separate contextual score notes, and a recording interval. Extra notes are anchored to neighbouring score events. Missed-note playback times are interpolated from surrounding aligned onsets, with a tempo-map estimate as fallback and an explicit unavailable state if timing cannot be estimated. Repetition labels include the replay interval, the first-pass interval, and the number of additional copies. The current stack uses timing to improve alignment but does not emit rhythm, intonation, or timbre-error labels. Those categories are excluded from the common evaluation scope.

4.7 Training and Configuration Selection

Synthetic generation supplies both the performed note sequence for acoustic training and its relationship to the clean score for alignment and error verification. Transcription targets describe the notes actually rendered, including wrong pitches, inserted notes, and repeat copies. This prevents the acoustic network from being trained to reproduce the intended score when the performance differs.

The current transcriber combines the CTC loss in Equation 1 with a weighted frame-level objective,

$$\mathcal{L}_{\text{acoustic}} = \mathcal{L}_{\text{CTC}} + 0.5\mathcal{L}_{\text{frame}}. \quad (4)$$

The frame objective supervises pitch, voicing, onsets, boundaries, rearticulations, and confidence, with additional within-note consistency terms. Feature-space augmentation varies gain, spectral balance, noise, pitch drift, articulation, reverberation-like tails, and brief transitions. These are training perturbations rather than a calibrated simulation of room acoustics. The selected checkpoint was fine-tuned on the repository’s realistic, fast-note, and practice-like synthetic training families; Appendix A.1 records its configuration.

The presence verifier is trained separately on played score notes and planted omissions. Extra- and missed-note classifiers use synthetic development predictions, including score-group-held-out calibration for the inherited gates. Deployment changes to gate thresholds and boundary rules are versioned separately from the acoustic checkpoint. The optional boundary forest uses archived synthetic development controls; those controls were reused during calibration and do not constitute an independent test. The acoustic model, verifier, and alignment procedure are therefore trained or selected in separate stages, without end-to-end gradient optimisation through the current dynamic program. The evaluation in Section 5 measures their combined output; component-level ablations are needed to attribute changes in performance.

5 Evaluation

We evaluate AudioEval with the dual-resolution CTC transcriber and the v9 passage aligner described in Section 4, using the fixed configuration in Appendix A.1. The comparison with Polytune and LadderSym covers synthetic clarinet recordings and 11 manually labelled recordings of human playing. The synthetic measure evaluates score correspondence and event labelling; the real-recording measure evaluates the type and timing of reported errors. These are end-to-end system results. Two case studies illustrate score alignment and feedback. The experimental v10 boundary refinement is outside the quantitative comparison.

Draft status: benchmark values in the comparison prose, tables, and plot are placeholders pending the current pipeline’s evaluation.

5.1 Experimental Setting

The comparison uses 358 synthetic validation recordings, all drawn from Mozart excerpts. Their reference annotations contain 40712 events, including correct notes, repeated notes, and errors. This set was available during development, and the reported values are validation results.

AudioEval uses the CTC checkpoint trained on the synthetic families described in Section 3.1, with checkpoint selection and alignment settings documented in Appendix A.1. We trained Polytune and LadderSym from scratch on a separate corpus of 10049 synthetic recordings and used 1116 recordings for model selection. The selected models completed 31 and 26 training epochs, respectively. Exact-audio comparisons identified 21 evaluation recordings in baseline training and 3 in baseline model selection. The systems share the evaluation set and scoring procedure, with the training and model-selection conditions described above.

All methods analyse each supplied evaluation clip in full, using the same recording boundaries. Polytune receives practice and reference audio; LadderSym additionally receives a symbolic score prompt. Their timestamped note predictions are assigned to written locations using the same conversion procedure. This procedure pairs compatible missed and extra predictions into wrong notes, aligns the remaining notes to the score, and checks for repeated passages. Its parameters are fixed for both baselines. Appendix A.2 provides the training and conversion details.

5.2 Evaluation Metrics

The primary measure is *score-based weighted micro-F1*. It assesses whether a predicted event is attached to the correct musical location and given the correct type. A written note is identified by its position in the score; during a repetition, its performed occurrence is also distinguished. An extra note has no written counterpart and is identified by its event in the generated performance.

Predictions and reference events are matched one-to-one. A pair receives credit 1 when both location and type agree, 0.5 when the location agrees but the type differs, and 0 when the location is incorrect or unavailable. Each event can contribute to at most one pair. If C is the total match credit, N_p the number of predicted events, and N_g the number of reference events, then

$$P = \frac{C}{N_p}, \quad R = \frac{C}{N_g}, \quad F_1 = \frac{2C}{N_p + N_g}. \quad (5)$$

The term *weighted* refers to the fractional match credit, and *micro* indicates that credit and event counts are pooled across recordings. Correct, repeated, and erroneous notes all contribute to the measure.

An event type includes its alignment relationship, note-content label, and repetition status. The common scoring scope excludes rhythm, intonation, and timbre for all systems; the current AudioEval exporter does not predict those categories. The same event-matching rule is applied to all three systems. Confidence intervals are estimated with 1000 bootstrap samples, resampling whole recordings to retain dependence among events in a performance.

5.3 Comparison with Existing Detectors

Table 2 and Figure 2 show the results. AudioEval achieved weighted micro-F1 of 61.04%, compared with 41.39% for Polytune and 34.71% for LadderSym.

Table 2: Score-based weighted precision, recall, and micro-F1 on 358 synthetic validation recordings. Values are percentages. Confidence intervals are estimated by resampling recordings 1000 times. *Provisional values; current evaluation pending.*

Method	Precision	Recall	F1	95% F1 interval
Polytune	44.27	38.85	41.39	[39.94, 42.74]
LadderSym	39.52	30.95	34.71	[33.25, 36.07]
AudioEval	55.52	67.79	61.04	[59.37, 62.72]

AudioEval achieved weighted precision of 55.52% and recall of 67.79%.

5.4 Recorded Clarinet Performances

The student author recorded 11 clarinet performances and manually reviewed their error labels. Table 3 reports precision, recall, and micro-F1 for AudioEval, Polytune, and LadderSym. A prediction matches an annotated error when their error types agree and their time intervals overlap sufficiently. We measure this overlap using interval intersection-over-union (IoU): the

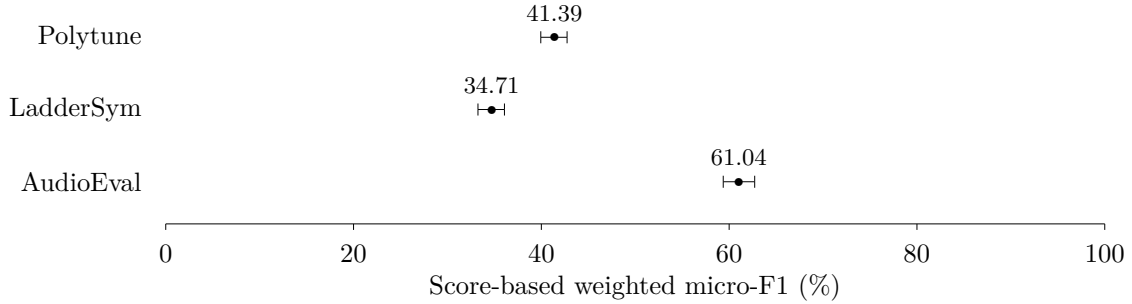


Figure 2: Score-based weighted micro-F1 for the three systems on the same 358 synthetic validation recordings. Markers show point estimates and horizontal lines show 95% recording-bootstrap confidence intervals. Training conditions and the conversion of baseline outputs are described in Section 5.1. *Provisional values; current evaluation pending.*

duration shared by the two intervals divided by their combined duration, counting the overlap once. A match requires IoU of at least 0.3, and each prediction and annotation can be matched only once.

This error-event measure counts only errors, whereas the synthetic measure also includes correctly played notes and gives partial credit for the correct score location. The two F1 measures therefore answer different questions and should be interpreted separately.

Table 3: Error-event precision, recall, and micro-F1 at interval $\text{IoU} \geq 0.3$ on 11 manually labelled clarinet recordings. Values are percentages. Only error labels are scored. *Provisional values; current evaluation pending.*

Method	Precision	Recall	F1
AudioEval	20.26	34.36	25.49
Polytune	1.03	18.31	1.96
LadderSym	1.78	15.49	3.20

The comparison uses the same trained Polytune and LadderSym models as the synthetic experiment. Their outputs are converted to the error categories used here. The scoring scope contains wrong notes, missed notes, extra notes, and unintended repetitions. Rhythm, intonation, and timbre annotations are excluded for every system. Reference annotations are reviewed and frozen before scoring; model predictions do not supply or overwrite the reference labels. Appendix A.5 gives the inference settings and scoring details.

5.5 Case Studies

These examples use saved v9 outputs for DataCreate samples 005 and 007. Sample 005 combines an earlier v9 alignment with the latest labels saved in the annotation interface, while sample 007 uses the passage-alignment revision. The author has checked the model labels in the second passage; older manual labels for that example are inaccurate and are not used as the reference. This targeted review does not establish test-set accuracy or complete error detection. Times are relative to each saved recording, and pitch names use written clarinet pitch.

Case 1: extra notes and a repeated bar. Sample 005 is a 38.14s clarinet recording covering bars 66–82. In the 9–16s window, the saved labels mark extra-note regions at 10.01–10.83s, 12.34–12.42s, and 13.14–13.32s. AudioEval links the replay at 13.32–15.26s to the eight-note score passage first attempted at 10.83–13.14s. The saved labels contain one further extra-note region outside this window.

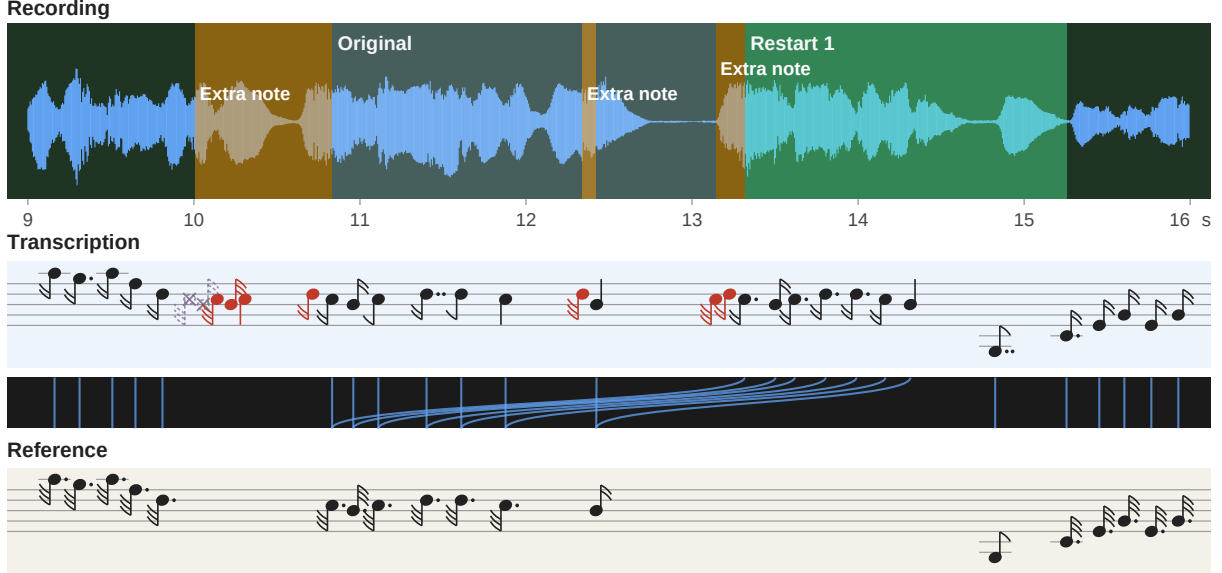


Figure 3: Sample 005 at 9–16s, showing the bar-70 repetition and adjacent extra-note labels. The upper staff shows transcription and the lower staff shows the reference score; lines connect corresponding notes. Red transcription notes indicate unmapped candidates. Coloured waveform regions show the latest labels saved in the annotation interface. “Original” and “Restart 1” identify the first attempt and its replay. The reference G3 is positioned at its first aligned occurrence, during the replay.

Figure 3 links both attempts to bar 70 before the alignment advances into bar 71. G3 is aligned only during the replay. Extra-note labels retain their recording intervals, while the repetition is shown as one passage-level event. Unmapped transcription candidates remain visible without automatically becoming separate extra-note labels.

Case 2: a repeated passage with note errors. Sample 007 is a 34.91s recording covering bars 114–130 of the Weber clarinet score. In bar 121, AudioEval reports C4 in place of the written D4 at 14.21s and an extra C4 at 14.69s. The player then returns to the beginning of that bar. AudioEval links the replay at 15.43–17.38s to the same 16 score notes as the original passage at 12.68–15.43s and reports one unintended repetition. After the replay, it identifies D4 in place of E4 in bar 122 at 17.51s. Across the complete clip, the saved feedback contains five wrong-note labels, one extra-note label, one missed-note label, and one repetition label. The missed note is D5 in bar 123, with an estimated interval at 20.43–20.56s, beyond the displayed window.

Figure 4 shows how the repeated notes reconnect to the earlier score passage while the subsequent notes continue into bar 122. The replay is presented as one passage-level event, without separate content-error labels inside that repeated attempt. The wrong and extra notes on the first attempt retain their own recording intervals and score locations. Together, the two examples illustrate how the current pipeline separates candidate inspection, score correspondence, and verified feedback.

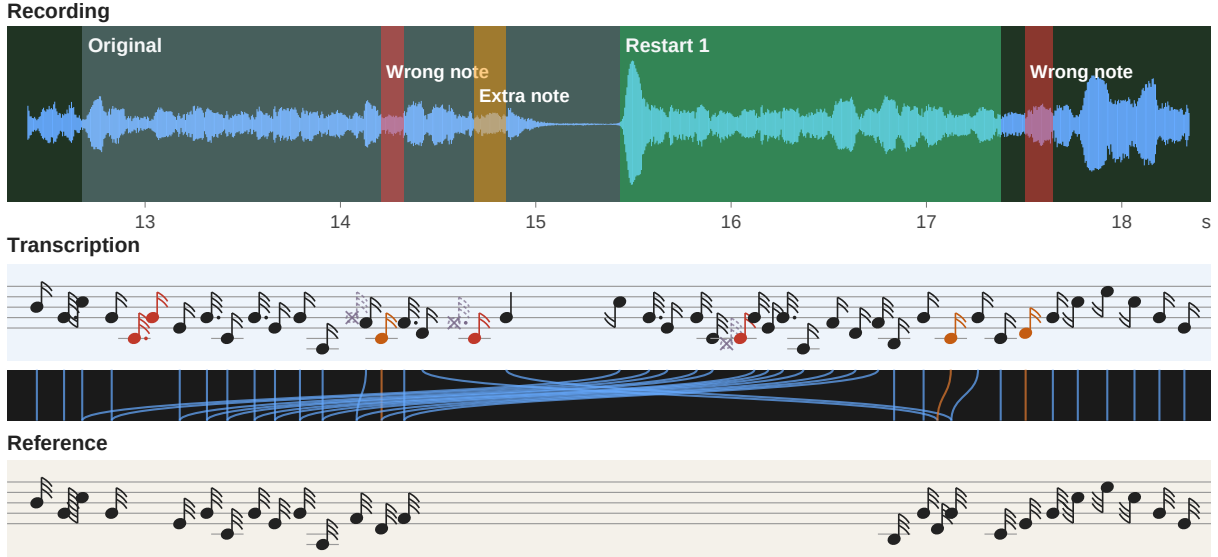


Figure 4: The bar-121 repetition and adjacent note errors in sample 007. The upper staff shows transcription and the lower staff shows the reference score; connecting lines associate performed notes with score locations. “Original” and “Restart 1” identify the first passage and its replay. Coloured waveform regions show emitted labels. Red or orange transcription notes show unmapped candidates or pitch substitutions, respectively; these do not all become separate error labels. The author confirmed the reported labels in this passage.

6 MusicPal: A Live Practice Companion and Data Flywheel

MusicPal is an implemented desktop practice companion that presents AudioEval’s feedback through score animation and audio playback. Figure 5 shows the physical device in use. Its saved-session workflow also provides a route to collect real clarinet performances for model development.

6.1 Implemented Device and Feedback Workflow

MusicPal handles recording, display, and playback, while a server runs AudioEval. The player selects a reference score and records or uploads a passage. AudioEval transcribes the take, retrieves and aligns the attempted passage, and identifies errors. The server retains the audio, reference score, alignment, and predicted labels alongside the generated feedback.

The feedback pipeline uses a language model and speech synthesis to explain the retained error labels. A score animation combines narration with reference and performed-note examples, highlights the affected region, and follows playback with a cursor. Performance notation is an approximate transcription for comparison with the written score. AudioEval supplies the error decisions; narration explains them without changing the labels. Uncertain alignments are withheld from assessment.

6.2 Growing a Clarinet Practice Dataset

Real clarinet practice recordings paired with intended scores and reviewed error annotations are a bottleneck for this project. Synthetic data cannot capture the full range of players, articulation, restarts, microphones, and room acoustics. Distributing MusicPal to students would let them receive targeted feedback while, with their permission, contributing practice sessions to a growing research collection.



Figure 5: MusicPal running score feedback, with notation, a highlighted note region, explanatory text, and playback controls. The still is cropped from the author’s demonstration video at 22.5 s.

The proposed data flywheel connects use, review, and model improvement. Each contributed session preserves its recording, score, model version, and predictions. An annotator listens, checks the score correspondence, and corrects note identities, error types, and locations, keeping reviewed labels separate from predictions. Reviewing uncertain and unflagged sessions alongside reported errors would also expose missed detections.

Reviewed training examples can complement synthetic data when updating the transcriber, aligner, and error-verification models. Evaluation recordings separated from training by player and source performance would test each update before release. Better feedback could encourage continued use and further contributions, completing the cycle. The live product makes this collection route practical; dataset growth and model gains remain to be measured as MusicPal reaches more players.

7 Conclusion

We present AudioEval, a layered method for clarinet practice feedback. The current method combines dual-resolution CTC transcription, bounded passage and repetition alignment, and acoustic verification of possible errors. It links note errors and repeated passages to canonical score locations and recording intervals, while retaining an explicit uncertain state. Same-pitch boundary restoration remains an experimental extension.

AudioEval achieved a score-based weighted micro-F1 of 61.04% on 358 synthetic validation recordings, compared with 41.39% for Polytune and 34.71% for LadderSym. Its weighted precision and recall were 55.52% and 67.79%, respectively. On 11 manually labelled clarinet recordings, AudioEval achieved an error-event micro-F1 of 25.49%, compared with 1.96% for Polytune and 3.20% for LadderSym. These results evaluate the CTC transcriber, passage aligner, and error verification together. Two case studies using saved v9 outputs illustrate the association of repetitions and note errors with score passages and recording intervals. The implemented MusicPal device presents audiovisual feedback and retains practice sessions. Distributing it to

more clarinet players can support a data flywheel: contributed recordings are reviewed, used to improve the models, and translated into better feedback for subsequent practice. The product demonstration establishes a working interface; the resulting dataset growth and model improvements remain future measurements.

Limitations

The evaluation measures the complete system; it does not isolate the contribution of the transcriber, passage aligner, or acoustic verification. The common label scope covers note-content errors and repetitions, so the results do not assess rhythm, intonation, or timbre. The optional v10 boundary refinement is outside the quantitative comparison. Same-pitch merging can erase real attacks; v10 can restore only already decoded, structurally supported boundaries and has been calibrated on reused development controls. Passage retrieval can choose the wrong occurrence of an identical phrase, and its earliest-plausible-location policy does not resolve that uncertainty acoustically. Bounded windows exclude long unsegmented performances and unrestricted jumps. Error gates trade recall for fewer false alarms, and their scores are not calibrated probabilities.

The synthetic comparison uses validation recordings under the training conditions reported in Section 5.1. The systems differ in training data and the labels they provide. The real-recording comparison covers 11 performances by the student author and uses an error-event measure, so its F1 values should be interpreted separately from the synthetic score-based results. The case studies use selected passages: sample 005 combines saved model alignment with labels edited in the annotation interface, while the labels discussed for sample 007 were checked by the author. Earlier inaccurate manual labels for sample 007 are not treated as ground truth. This review does not establish completeness of detection or aggregate accuracy. Evaluation across more players, pieces, and recording conditions, with independent annotations and separate development and evaluation data, would establish how broadly the results apply. Further experiments could vary transcription, score alignment, and error labelling one component at a time on the same data.

MusicPal is operational, but the demonstration does not establish usability, learning gains, or reliability across players and recording conditions. Further tests should measure capture quality, feedback latency, repeated-session reliability, and whether users find the feedback useful during practice. The proposed data flywheel also depends on continued participation and annotation effort. Contributed recordings may overrepresent particular players, pieces, or devices; reviewing diverse sessions and keeping evaluation recordings separate from training will be necessary to assess broader improvements. Dataset growth and model gains from deployment have not yet been quantified.

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A Implementation and Evaluation Details

A.1 Current Transcriber and Aligner Configuration

The following settings define the AudioEval configuration used in the quantitative comparison and case studies. The selected acoustic checkpoint is `precision-v4/v5-dclike/best.pt`; v9 and v10 reuse this checkpoint rather than the later pilot transcriber. It uses 32 spectral channels, temporal width 256, twelve dilated blocks, and dropout 0.10. The long mel branch covers 45–8,000 Hz and the short branch 100–8,000 Hz. Inference uses 4,096-frame windows with 1,024-frame overlap.

The final fine-tuning stage starts from `precision-v4/v4-realism/best.pt` and combines the `realistic92`, `fast102`, and `dclike11` training caches. It runs three epochs with batch size eight, 2,048-frame crops, AdamW weight decay 10^{-3} , and a one-cycle schedule peaking at 1.5×10^{-4} . Gradient norm is clipped at 2.0; the seed is 20260927. The selected checkpoint is from epoch two, using the recorded validation selection score. These are the final fine-tuning settings, not the full pretraining history.

The default runtime is `v9-passage-v2`, using the v9 candidate’s decoder and same-pitch settings and the bounded v5 aligner. It inherits the v6 acoustic verifier and logistic gate coefficients, with the missed-note threshold changed to 0.80; the extra-note threshold remains 0.85. Table 4 records the main operating parameters. Hashes in the candidate manifests identify the checkpoint and versioned components.

Table 4: Current decoding and alignment settings. These are operating parameters, not measured accuracies.

Component	Setting
CTC candidates	Blank scale 0.5; optional peak threshold 0.08; alternative-pitch support at least 0.10.
Transition candidates	At most 45 ms to the next primary onset and attack support below 0.35.
Passage retrieval	Reliable-note confidence at least 0.20; at least eight reliable notes for long-score retrieval; similarity at least 0.52. Scores with at most 128 notes bypass retrieval.
Passage ranking	Alignment cost per candidate plus $0.4(1 - \text{similarity})$. Earliest plausible location within similarity difference 0.04 and rank difference 0.08 of the best.
Search bounds	Four windows; 512 score notes per window; 1,024 repeat hypotheses per window; 2,048 expanded units and two million DP cells per hypothesis; 20 million locator cells.
Base edit costs	Substitution 1.1; insertion scale 1.2; deletion 1.0; merge 0.15; repeat-copy penalty 1.0; alternative-pitch penalty 0.6.
Timing refinement	Anchor confidence 0.90; $\lambda_t = 0.2$, $\kappa = 1$, and $\tau_j = \max(0.06 \text{ s}, 0.35 \hat{d}_j)$ for expected duration $\hat{d}_j$.
Error verification	Extra threshold 0.85; missed threshold 0.80; whole-clip match/copy fraction at least 0.45. Boundary-context notes outside audible coverage are not automatically labelled missed.

The v9 same-pitch rule examines up to 90 ms on each side, clipped at neighbouring onsets with a 5 ms margin. Its gap veto uses an 18 dB depth and a configured 10 ms interruption target. The observed low-energy core is shorter because the RMS window spans 4 ms; the rule compensates for this window and sampling phase rather than treating the core as an exact silence-duration estimate. The permissive energy-range limit is 54 dB, with normalised mel step and drift bounds of 0.35 and 0.65; model attack and voicing vetoes are disabled in this candidate. These settings were calibrated using injected split controls and are known to merge

genuine repeated notes.

The separate **v10-boundary-v1** configuration adds a 96-tree boundary forest with maximum depth five, minimum leaf size three, and seed 20261005. It requires original primary-candidate confidence at least 0.5 and a repeat vote at least 0.5, followed by the realignment checks in Section 4.3. Its 442 usable archived acoustic rows come from synthetic development controls. The control material was reused during calibration, including exploratory inspection of the check subset; neither the control replay nor unchanged sparse error labels establishes independent boundary accuracy.

A.2 Baseline Training

Polytune and LadderSym are trained from scratch on 10049 synthetic recordings, with 1116 recordings for model selection. The models used in the comparison completed 31 and 26 epochs, respectively. Both use seed 365, peak learning rate 2×10^{-5} , 4,000 warmup updates, a cosine schedule configured for 40 epochs, and gradient accumulation over 16 training steps. Training samples 2.048-second audio segments; evaluation covers complete recordings.

Inference uses 16 kHz mono audio, 32-bit floating-point computation, greedy decoding, and a maximum of 1,024 output tokens. LadderSym additionally receives a reference-score prompt of up to 1,024 tokens. The same 358 evaluation recordings and 40712 reference events are used for all three systems. Exact-audio overlap with baseline training and model selection is reported in Section 5.1.

A.3 Assigning Baseline Predictions to Score Locations

The baselines produce timestamped notes with the native labels **Correct**, **Missing**, and **Extra**. We use *missed note* in the paper for the category denoted by **Missing**. The same procedure assigns both baselines' predictions to score locations before evaluation.

An extra-note prediction and a missed-note prediction form a wrong-note prediction when their pitches differ, they are mutually nearest in onset time, and their onsets are within 100 ms. Correct notes, these paired predictions, and remaining missed notes are then aligned monotonically to the reference score. The edit costs are 1 for insertion or deletion and 1.5 for a pitch mismatch. A paired wrong-note prediction uses the expected pitch of the missed note to determine its score location. Gaps introduced by conversion do not create additional missed-note predictions.

A remaining sequence of at least three extra notes may be assigned to a repetition if its pitches agree with at least 80% of a previously visited contiguous score sequence. The longest qualifying sequence is selected; ties favour pitch agreement and then the most recent score position. Previous predicted visits determine the performed occurrence assigned to the repeated notes.

The conversion receives only model predictions and the reference score. Its parameters were fixed before scoring. Unlocated and duplicate predictions remain in the prediction count, and no performance-event identity is assigned without evidence from the prediction. Rhythm, intonation, and timbre are excluded from event typing for every system, matching the common scoring scope in the main text.

A.4 Scoring and Confidence Intervals

Each system's predictions are saved before evaluation annotations are read. The one-to-one matcher selects the assignment with maximum total credit under the location and type rules in Section 5. Credit and event counts are pooled across recordings before precision, recall, and F1 are calculated.

Confidence intervals use 1000 bootstrap replicates. Each replicate samples 358 recordings with replacement and recomputes the pooled weighted micro-F1. The interval endpoints are the 2.5th and 97.5th percentiles of the resulting values. Resampling recordings preserves the grouping of notes within a performance. This gives the 95% interval reported in the table and figure.

A.5 Real-Recording Evaluation

The real-recording comparison uses the previously selected Polytune and LadderSym models after 31 and 26 training epochs, respectively. Both analyse complete recordings with 16 kHz mono inputs and bfloat16, a 16-bit floating-point format. Recordings are processed one at a time, with a maximum of 1,024 output tokens. LadderSym also receives a fixed symbolic encoding of the reference score. Predicted error intervals are evaluated individually, without merging adjacent intervals.

The annotations contain 71 scored error events across the 11 recordings. Matching requires the same error type and interval IoU of at least 0.3, with one-to-one assignment. Each match receives one credit; no partial credit or separate pitch-matching condition is applied. Precision, recall, and micro-F1 are computed from counts pooled across recordings. Polytune matches 13 reference events with 1258 predictions; LadderSym matches 11 with 617 predictions. Only wrong-note, missed-note, extra-note, and repetition events enter these counts; rhythm, intonation, and timbre annotations are excluded for every system.